

Homework 7: Robust Regression

Submission Notes: Please write your answers to each question under that question (not altogether at the end). It will be easier for me to give full credit if your answer is commented well, including comments about where you are completing each sub-part of the question. For an example, see Homework 1.

1. Below, I simulate data with heteroskedastic error, noting the variance is a function of the predictor x .

```
set.seed(1933)
x <- 1:10
(beta0 <- runif(1, -5, 5))

## [1] 0.08868237

(beta1 <- runif(1, -5, 5))

## [1] -2.116944

y <- beta0 + beta1 * x + rnorm(10, mean = 0, sd = x) # Variance is  $x^2$ 
dat <- tibble(x = x,
              y = y)
```

When I fit an OLS model, it doesn't "pick up" the regression equation.

```
mod.ols <- lm(y~x, data=dat)
summary(mod.ols)$coefficients

##               Estimate Std. Error   t value   Pr(>|t|)
## (Intercept) -6.4290175  2.8929497 -2.222305 0.05697684
## x           -0.7669293  0.4662411 -1.644920 0.13860636
```

In Figure 1, we see that the increased variance almost creates a quadratic (if we ignore the observation at $x = 10$)

```
ggplot(data=dat)+
  geom_point(aes(x=x, y=y))+
  theme_bw() +
  geom_abline(aes(intercept=coef(mod.ols)[1],
                  slope=coef(mod.ols)[2],
                  color="OLS"))+
  geom_abline(aes(intercept=beta0,
                  slope=beta1,
                  color="Truth"))+
  labs(color="")
```

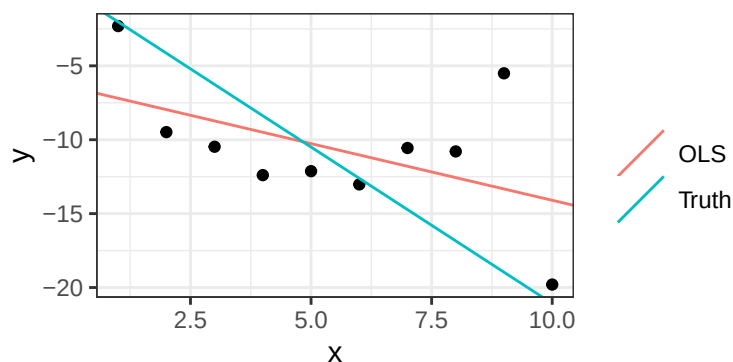


Figure 1: Heteroskedastic regression data generated for Question 1.

- (a) Conduct a residual and influence analysis on the `mod.ols`.

Note: The *known* issue may not be obvious.

```
source("plotResiduals.R")
source("plotInfluence.R")

plotResiduals(mod.ols)
```

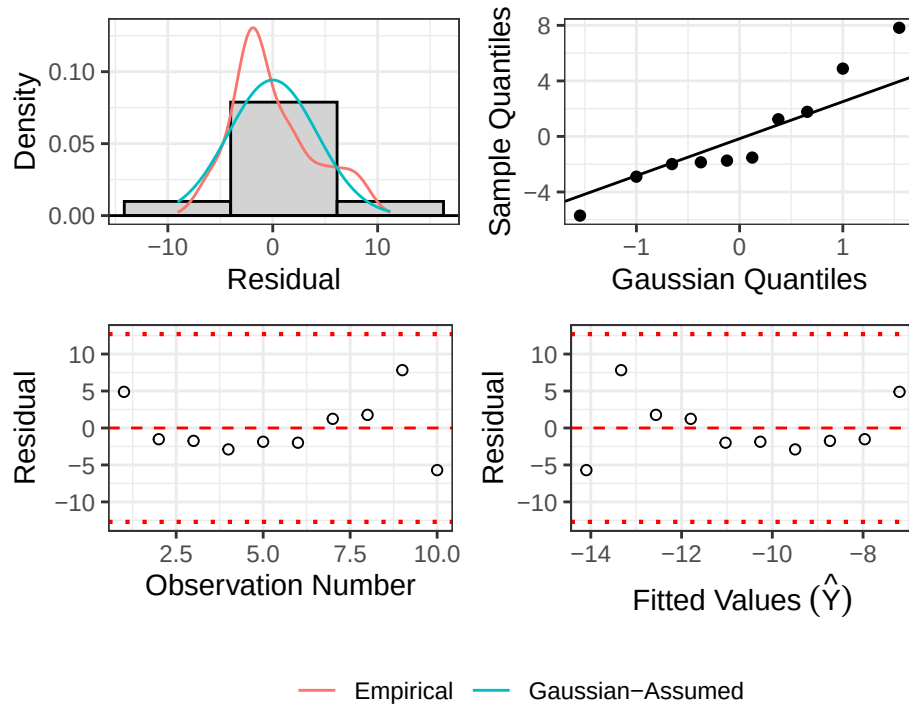


Figure 2: Residual Analysis on `mod.ols`

If I was analyzing the residuals of this model I would assume there was a non-linear relationship between the predictors and the response variable due to the clear curved pattern in the residuals. The histogram of the residuals looks a little skewed but it's not terrible for 10 observations. The qqplot indicates a potential non-linear relationship with the quadratic looking pattern on the right tail.

```
plotInfluence(mod.ols)
```

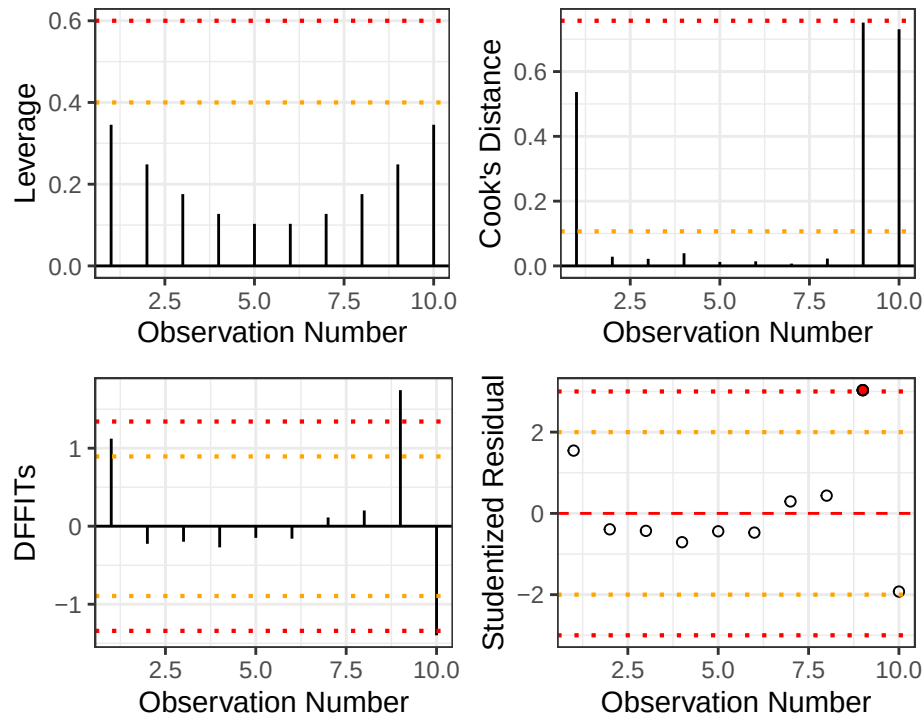


Figure 3: Influence Analysis on `mod.ols`

Looking at the influence, the thing that stands out is the clear leverage pattern across observation numbers, which would indicate to me that the leverage is dependent on x . The studentized residuals show similar problems to the residuals in Figure: 1a. Additionally, with there only being 10 observations, several of them having large Cook's d and DFFITS is a bit concerning due to the sheer proportion of potentially high influence points making up a large part of the model.

(b) Fit a weighted least squares (WLS) regression model.

i. Why is $w_i = 1/x_i^2$ a sensible choice for the weights?

Hint: Why is the RMSE ≈ 1 ?

In the setup of the regression equation the variance at each point $x(1:10)$ is x^2 , so weighting each point by $\frac{1}{x^2}$ should make each point have an equal weighted variance of 1.

ii. Fit the model with these weights, how does the result change?

```
weights = 1/x^2

mod_weighted <- lm(y ~ x, data = dat, weights = weights)

(sum <- summary(mod_weighted))

##
## Call:
## lm(formula = y ~ x, data = dat, weights = weights)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.0210 -0.6560  0.0217  0.7014  1.4473
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.7035      1.3762  -1.238  0.25088
## x             -1.8698      0.5418  -3.451  0.00868 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 1.145 on 8 degrees of freedom
## Multiple R-squared:  0.5982, Adjusted R-squared:  0.548
## F-statistic: 11.91 on 1 and 8 DF,  p-value: 0.008678
```

Now that we used the weighted model, the coefficient β_1 associated with x is estimated to be -1.8698 ($p = 0.00868$). This is only a few tenths greater than the true value of $\beta_1 = -2.1169$. The intercept β_0 is estimated to be -1.7035 while the true β_0 is 0.0887 , indicating that the estimate for the intercept is still quite off. This would probably be better if the data extended more for $x \leq 0$.

iii. Add the WLS line to the scatterplot.

```
betas <- sum$coefficients
beta0_w <- betas[1,1]
beta1_w <- betas[2,1]

ggplot(data=dat)+
  geom_point(aes(x=x, y=y))+
  theme_bw() +
  geom_abline(aes(intercept=coef(mod.ols)[1],
                  slope=coef(mod.ols)[2],
                  color="OLS"))+
  geom_abline(aes(intercept=beta0,
                  slope=beta1,
                  color="Truth"))+
  geom_abline(aes(intercept = beta0_w,
                  slope=beta1_w,
                  color="WLS")) +
  labs(color="") +
  geom_hline(yintercept = 0, linetype="dashed")
```

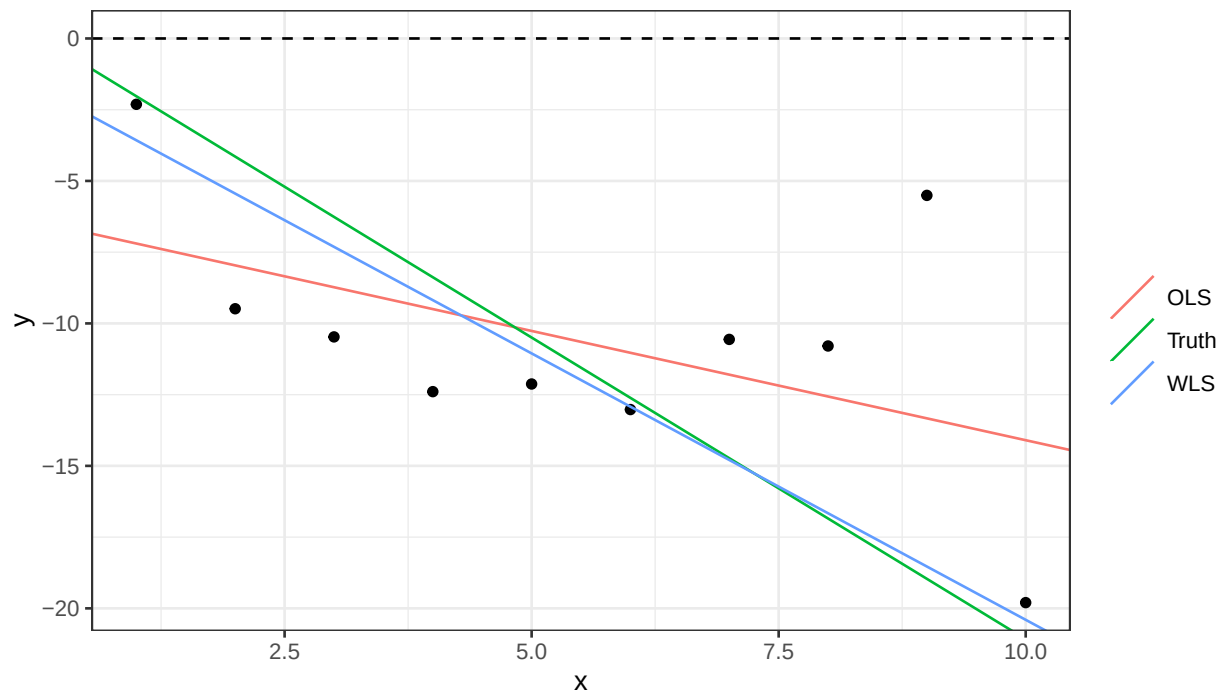


Figure 4: Scatterplot with Estimated Regression Lines (Updated With Weighted OLS Model)

The weighted model worked as intended here, as clearly the WLS regression line is much closer to the truth than the OLS regression line.

2. Explore fitting a regression model to the data simulated below.

```

n <- 50
set.seed(7272)
x <- sample(x=seq(0,100,0.01), size=n, replace=TRUE)
e <- rnorm(n=50, mean=0, sd=5)
y <- 3.5 + 2.1*x + e

first_dat <- tibble(x = x,
                    y = y)

# Add a misbehaved point
x2 <- c(x, 100)
y2 <- c(y, 25)

```

- (a) Fit an OLS model based on x and y . Check the assumptions of the model and interpret the coefficients of the model; are they what you would expect?

```
first_mod <- lm(y ~ x, data = first_dat)
```

```

first_mod <- lm(y ~ x, data = first_dat)
plotResiduals(first_mod)

```

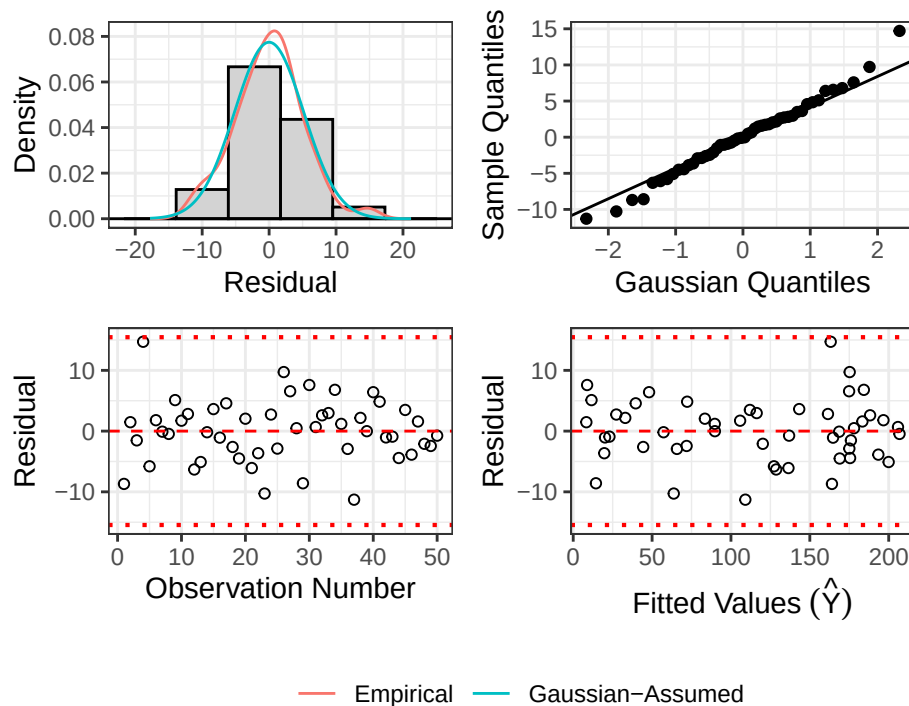


Figure 5: Residual Analysis on x and y OLS model

The residuals look perfect. They're centered at 0 with good normality. The qqplot is very reasonable. Finally, the residuals are indiscriminantly scattered with constant variance.

```
plotInfluence(first_mod)
```

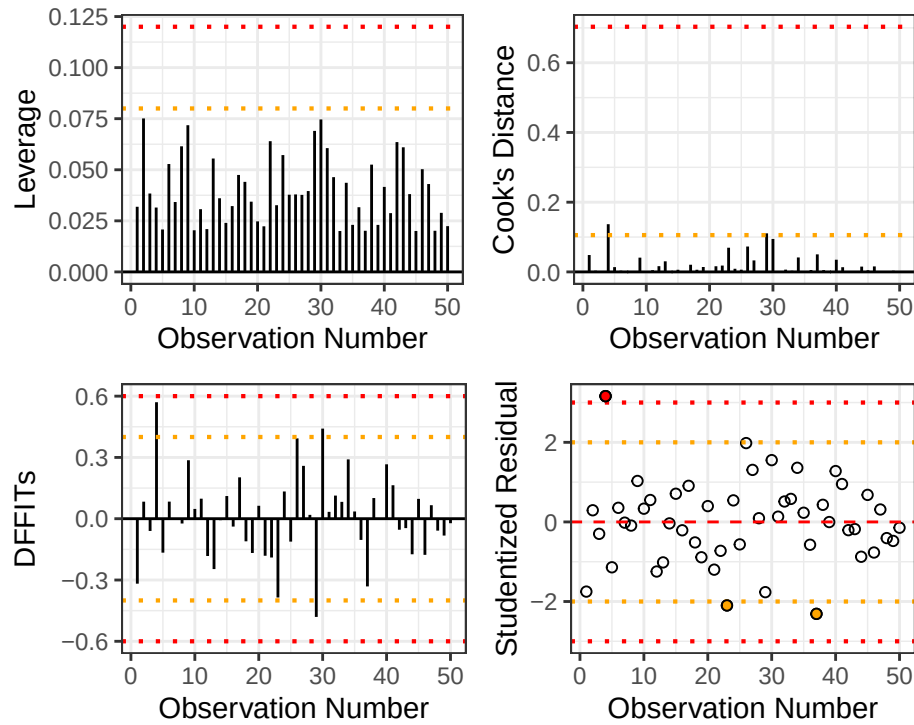


Figure 6: Influence Analysis on x and y OLS model

The influence analysis on the OLS model looks good as well, with reasonable leverage, DFFITS and Cook's d, with only one point having a potentially unusually high studentized residual (not unusual even for normal, randomized ϵ 's).

```
library(xtable)
library(effects)

first_sum <- data.frame(summary(first_mod)$coefficients)

colnames(first_sum) <- c("Estimate", "SE", "t", "p-value")
rownames(first_sum) <- c("Intercept", "x")

first_sum$'p-value' <- ifelse(first_sum$'p-value' < 0.001, "<0.001", first_sum$'p-value')

first_sum <- xtable(first_sum,
  label = "first.model.sum",
  caption = "OLS Model Summary (Without Outlier)",
  digits = 3)
```

	Estimate	SE	t	p-value
Intercept	5.223	1.443	3.620	<0.001
x	2.061	0.023	87.888	<0.001

Table 1: OLS Model Summary (Without Outlier)

```
ggplot(data = first_dat, aes(x = x, y = y)) +
  geom_point() +
  geom_abline(aes(intercept = first_sum$Estimate[1],
                  slope = first_sum$Estimate[2],
                  color = "OLS Estimate")) +
  geom_abline(aes(intercept = 3.5,
                  slope = 2.1,
                  color = "True Line")) +
  theme_bw() +
  labs(color = "") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  ggtitle(bquote(
    "OLS estimate: " ~
    .(round(first_sum$Estimate[1], 3)) +
    .(round(first_sum$Estimate[2], 3)) * x
  ))
))
```

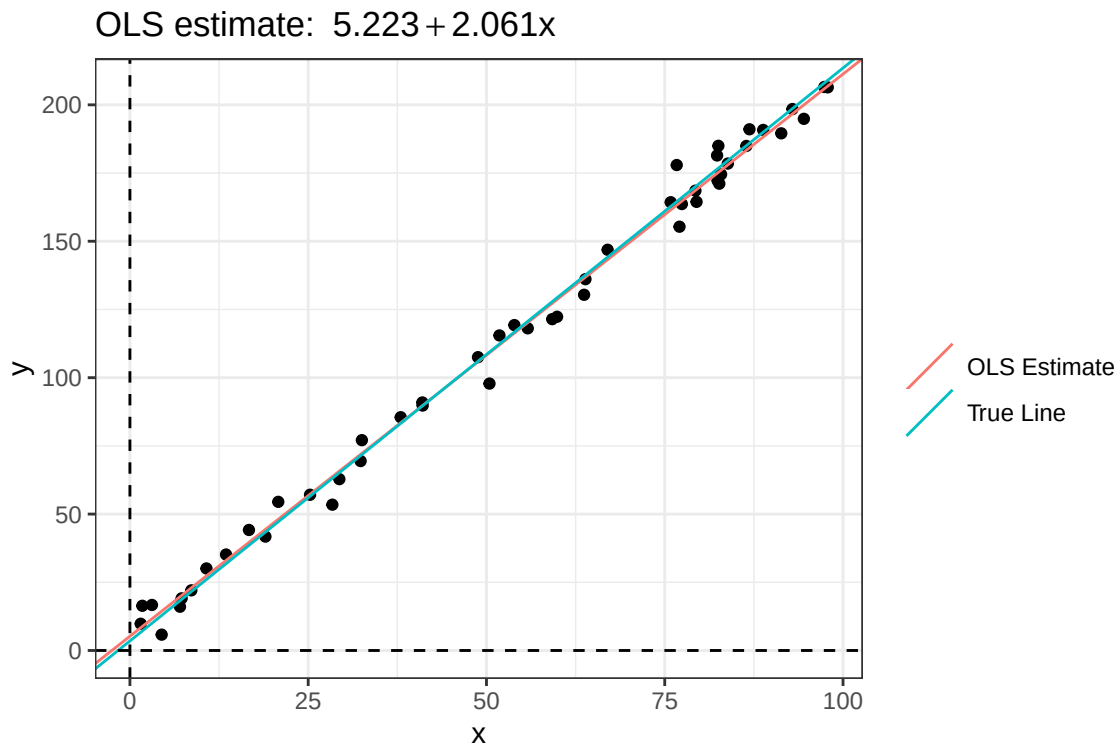


Figure 7: Scatterplot of the Data with the OLS Regression Line

The coefficient for x is very close to the true value. The intercept is reasonably close when looking at the plot. Overall, the two lines are almost visually indistinguishable.

- (b) Fit an OLS model based on x_2 and y_2 . Check the assumptions of the model and interpret the coefficients of the model; are they what you would expect?

```
second_dat <- tibble(x = x2,
                    y = y2)

second_mod <- lm(y ~ x, data = second_dat)
```

```
plotResiduals(second_mod)
```

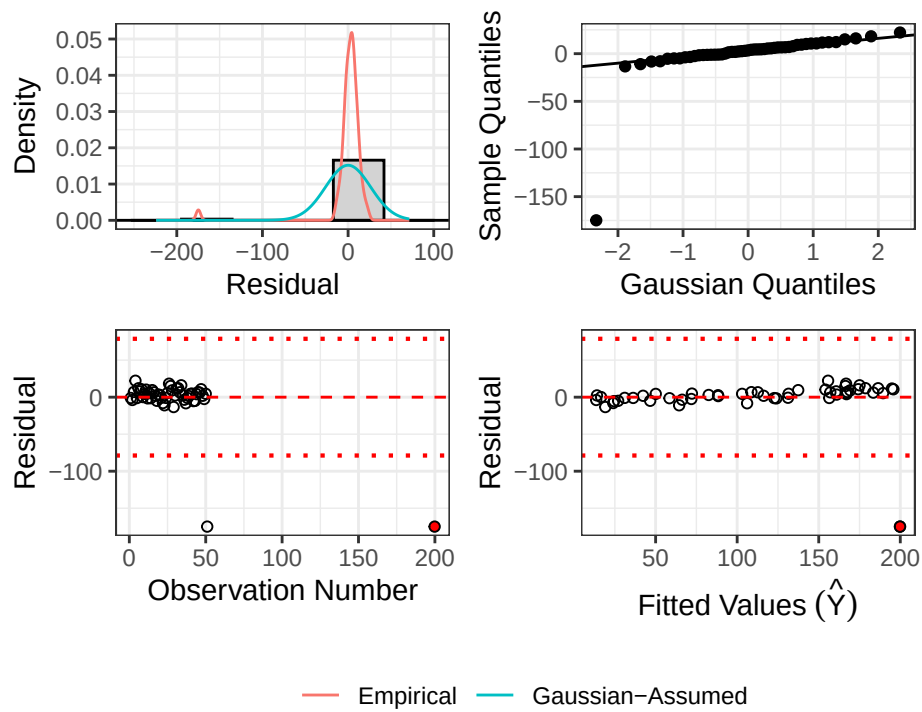


Figure 8: Residual Analysis on x_2 and 2 OLS model

The residuals are pretty hard to analyze here. They're centered around 0 with pretty good normality, just with that huge left tail. We can also see in the qqplot that the data fits perfectly except for the most negative quantile. If we don't focus on the huge outlier, I think the residuals look homoskedastic.


```
plotInfluence(second_mod)
```

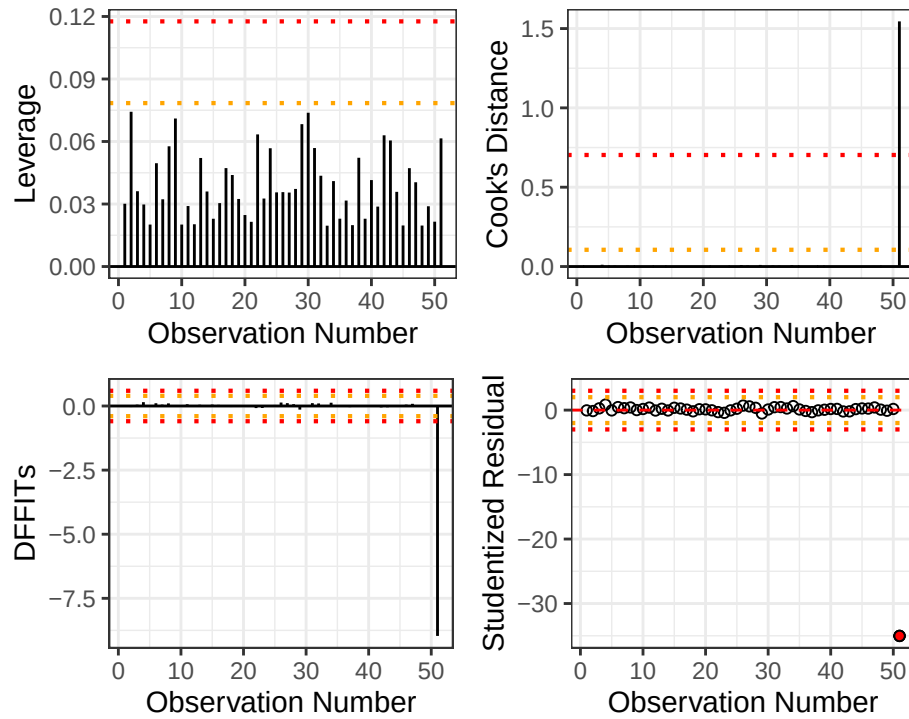


Figure 9: Influence Analysis on x2 and y2 OLS model

Again, it's quite hard to see everything but besides the huge outlier we don't see any other points on the boundaries of Cook's d, DFFITS, and the studentized residuals. Of course, the x value for our outlier isn't strange so there are no points with concerning leverage.

```
second_sum <- data.frame(summary(second_mod)$coefficients)

colnames(second_sum) <- c("Estimate", "SE", "t", "p-value")
rownames(second_sum) <- c("Intercept", "x")

second_sum$'p-value' <- ifelse(second_sum$'p-value' < 0.001, "<0.001", round(second_sum$'p-value',3))

second_sum <- xtable(second_sum,
  label = "second.model.sum",
  caption = "OLS Model Summary (With Outlier)",
  digits = 3)
```

	Estimate	SE	t	p-value
Intercept	10.742	7.314	1.469	0.148
x	1.891	0.117	16.160	<0.001

Table 2: OLS Model Summary (With Outlier)

```

ggplot(data = second_dat, aes(x = x, y = y)) +
  geom_point() +
  geom_abline(aes(intercept = second_sum$Estimate[1],
                  slope = second_sum$Estimate[2],
                  color = "OLS Estimate")) +
  geom_abline(aes(intercept = 3.5,
                  slope = 2.1,
                  color = "True Line")) +
  theme_bw() +
  labs(color = "") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  ggtitle(bquote(
    "OLS estimate: " ~
    .(round(second_sum$Estimate[1], 3)) +
    .(round(second_sum$Estimate[2], 3)) * x
  ))

```

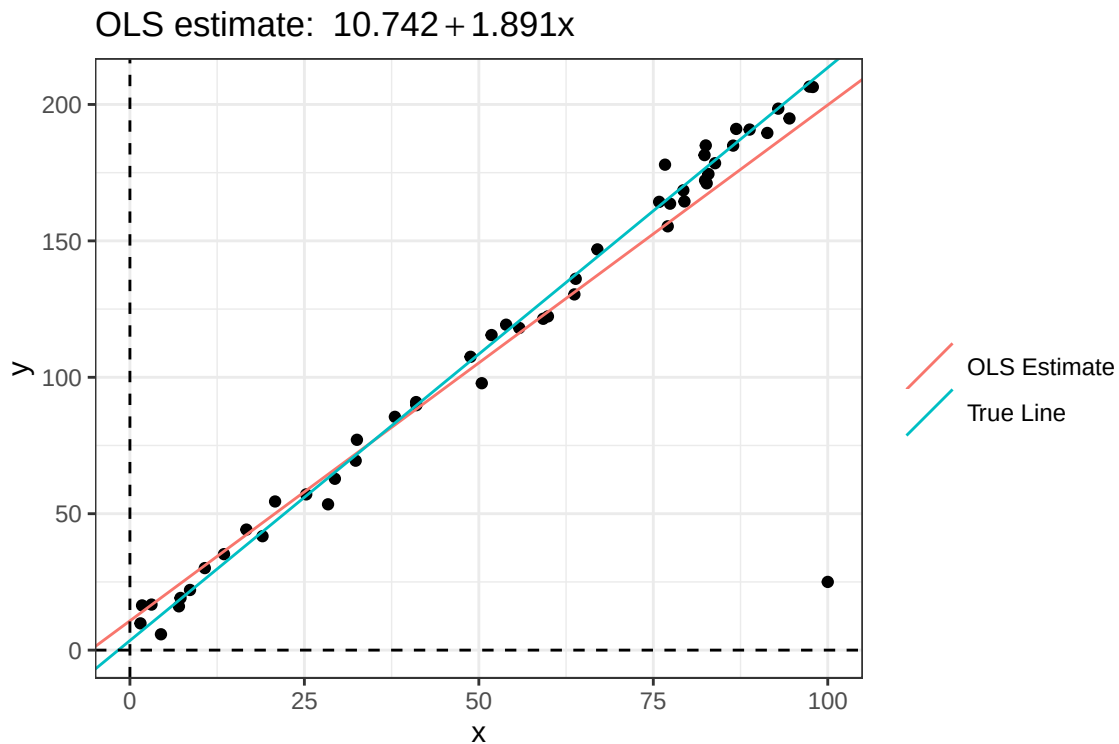


Figure 10: Scatterplot of the Data with the OLS Regression Line

Now that the outlier point is included in the model, the β_1 estimate for x goes from 2.061 to 1.891, pulling it from the true value of 2.1. Also, the β_0 estimate of the intercept goes from 5.223 to 10.742, more than doubling and sliding further from the true value of 3.5. We can see those difference in the plot, as the OLS estimate has been dragged down by the outlying point and that makes the OLS estimate worse and worse as x increases.

3. Continue with Question 2.

- (a) Fit an IRWLS model based on x_2 and y_2 using Huber weights. Interpret the coefficients of the model; are they what you would expect?

```

IWH_mod <- MASS::rlm(y ~ x, data = second_dat, psi = 'psi.huber')
IWH_sum <- data.frame(summary(IWH_mod)$coefficients)

colnames(IWH_sum) <- c("Estimate", "SE", "t")
rownames(IWH_sum) <- c("Intercept", "x")

```

```
IWH_sum <- xtable(IWH_sum,
  label = "IWH.model.sum",
  caption = "IRLWS Model Summary (With Outlier and Huber Weights)",
  digits = 3)
```

	Estimate	SE	t
Intercept	6.039	1.422	4.246
x	2.042	0.023	89.747

Table 3: IRLWS Model Summary (With Outlier and Huber Weights)

```
ggplot(data = second_dat, aes(x = x, y = y)) +
  geom_point() +
  geom_abline(aes(intercept = IWH_sum$Estimate[1],
    slope = IWH_sum$Estimate[2],
    color = "Huber IRWLS Estimate")) +
  geom_abline(aes(intercept = 3.5,
    slope = 2.1,
    color = "True Line")) +
  theme_bw() +
  labs(color = "") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  ggtitle(bquote(
    "Huber IRWLS estimate: " ~
    .(round(IWH_sum$Estimate[1], 3)) +
    .(round(IWH_sum$Estimate[2], 3)) * x
  ))
```

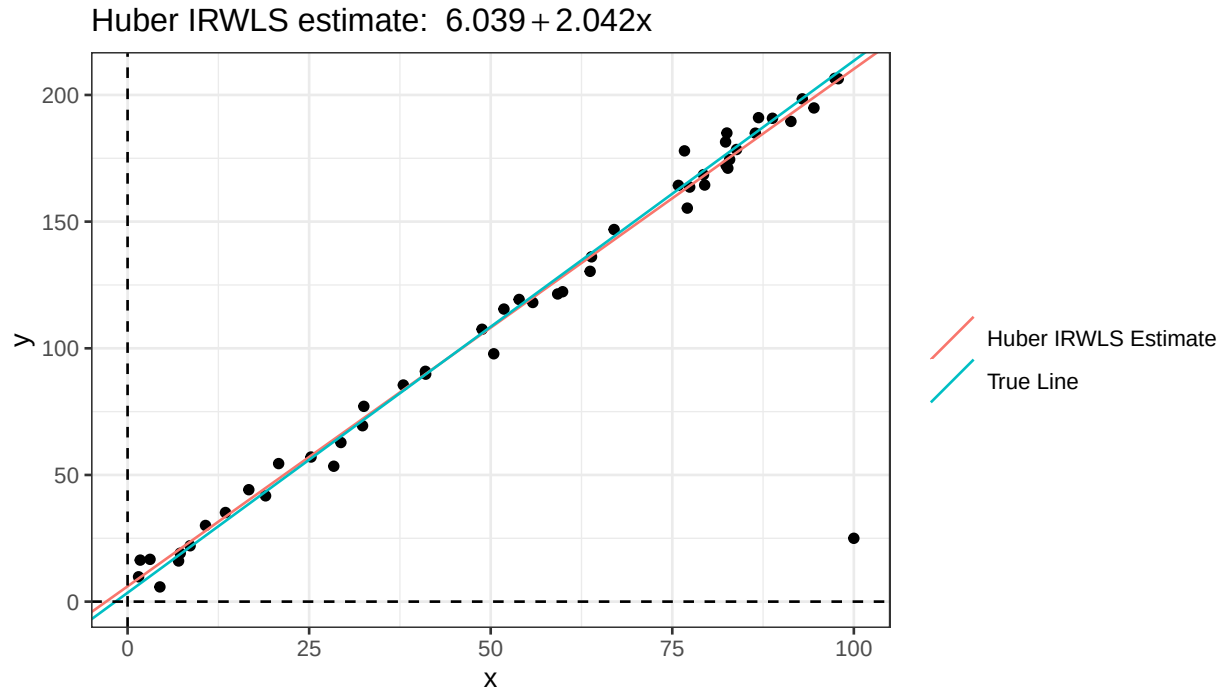


Figure 11: Scatterplot of the Data with the IRWLS Regression Line (Huber Weights)

The IRWLS model with Huber weights β 's are much closer to the true value than the OLS model. The IRWLS with Huber weights estimate for β_0 is 6.039 versus the 10.742 of the OLS (true value is 3.5). Also, the β_1 is 2.042 versus the 1.891 in the OLS model (true value is 2.1). So while

the regression is still being pulled down slightly by the outlier, it's noticeable better performing empirically and visually than the standard OLS model.

- (b) Fit an IRWLS model based on x_2 and y_2 using Bisquare weights. Interpret the coefficients of the model; are they what you would expect?

```
IWB_mod <- MASS::rlm(y ~ x, data = second_dat, psi = "psi.bisquare")

IWB_sum <- data.frame(summary(IWB_mod)$coefficients)

colnames(IWB_sum) <- c("Estimate", "SE", "t")
rownames(IWB_sum) <- c("Intercept", "x")

IWB_sum <- xtable(IWB_sum,
  label = "IWB.model.sum",
  caption = "IRLWS Model Summary (With Outlier and Bisquare Weights)",
  digits = 3)
```

	Estimate	SE	t
Intercept	5.702	1.434	3.976
x	2.050	0.023	89.372

Table 4: IRLWS Model Summary (With Outlier and Bisquare Weights)

```
ggplot(data = second_dat, aes(x = x, y = y)) +
  geom_point() +
  geom_abline(aes(intercept = IWB_sum$Estimate[1],
                  slope = IWB_sum$Estimate[2],
                  color = "Bisquare IRWLS Estimate")) +
  geom_abline(aes(intercept = 3.5,
                  slope = 2.1,
                  color = "True Line")) +
  theme_bw() +
  labs(color = "") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  ggtitle(bquote(
    "Bisquare IRWLS estimate: " ~
    .(round(IWB_sum$Estimate[1], 3)) +
    .(round(IWB_sum$Estimate[2], 3)) * x
  ))
))
```

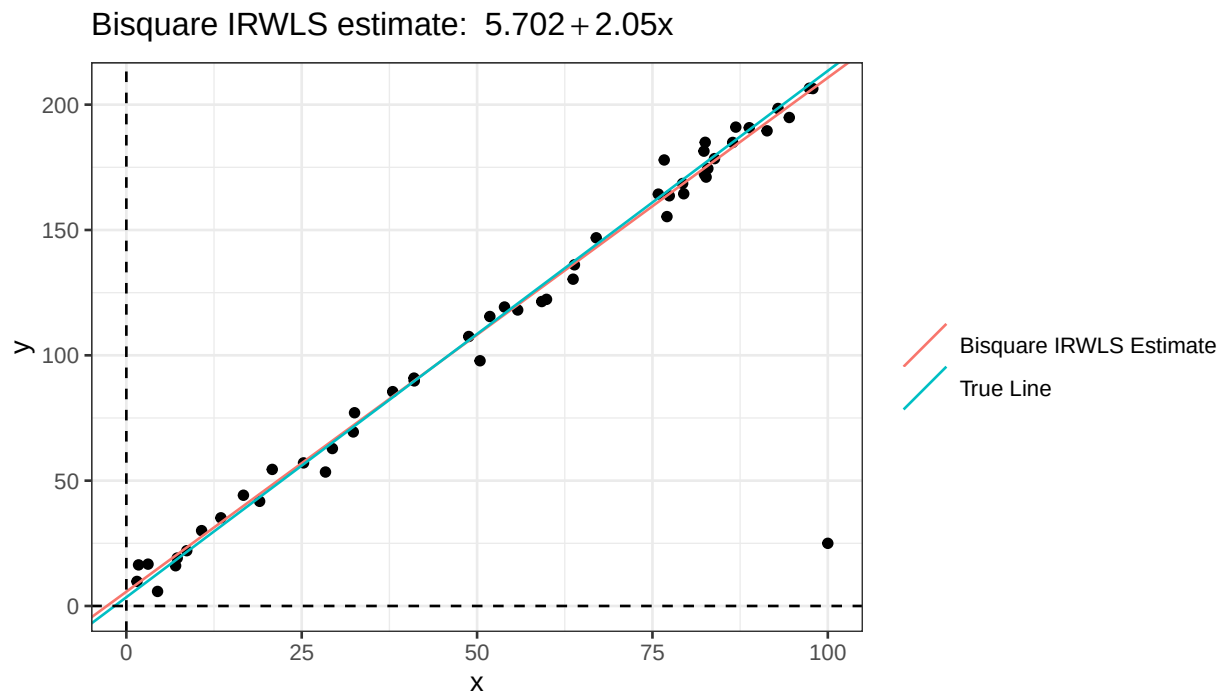


Figure 12: Scatterplot of the Data with the IRWLS Regression Line (Bisquare Weights)

The IWRLS model with Bisquare β 's are also very close to the true values and are very close to the Huber weighted IRWLS. The β_0 estimate is 5.702 compared to the Huber estimate of 6.039 and the true value of 3.5. The β_1 estimate is 2.050 compared to the Huber estimate of 2.042 and the true value of 2.1. So in this case, the Bisquare estimates are a tiny bit closer than the Huber estimate.

- (c) Fit an IRWLS model based on x_2 and y_2 using `lmrob()` from the `robustbase` package for R. Interpret the coefficients of the model; are they what you would expect?

```
rob_mod <- robustbase::lmrob(y ~ x, data = second_dat)

rob_sum <- data.frame(summary(rob_mod)$coefficients)

colnames(rob_sum) <- c("Estimate", "SE", "t", "p-value")
rownames(rob_sum) <- c("Intercept", "x")

rob_sum$p-value <- ifelse(rob_sum$p-value < 0.001, "<0.001", round(rob_sum$p-value, 3))

rob_sum <- xtable(rob_sum,
  label = "rob.model.sum",
```

```
caption = "IRLWS Model Summary (With Outlier and RobustBase Package)",
digits = 3)
```

	Estimate	SE	t	p-value
Intercept	5.633	1.382	4.077	<0.001
x	2.052	0.022	93.266	<0.001

Table 5: IRLWS Model Summary (With Outlier and RobustBase Package)

```
ggplot(data = second_dat, aes(x = x, y = y)) +
  geom_point() +
  geom_abline(aes(intercept = rob_sum$Estimate[1],
                  slope = rob_sum$Estimate[2],
                  color = "RobustBase IRWLS Estimate")) +
  geom_abline(aes(intercept = 3.5,
                  slope = 2.1,
                  color = "True Line")) +
  theme_bw() +
  labs(color = "") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  ggtitle(bquote(
    "RobustBase IRWLS estimate: " ~
    .(round(rob_sum$Estimate[1], 3)) +
    .(round(rob_sum$Estimate[2], 3)) * x
  ))
))
```

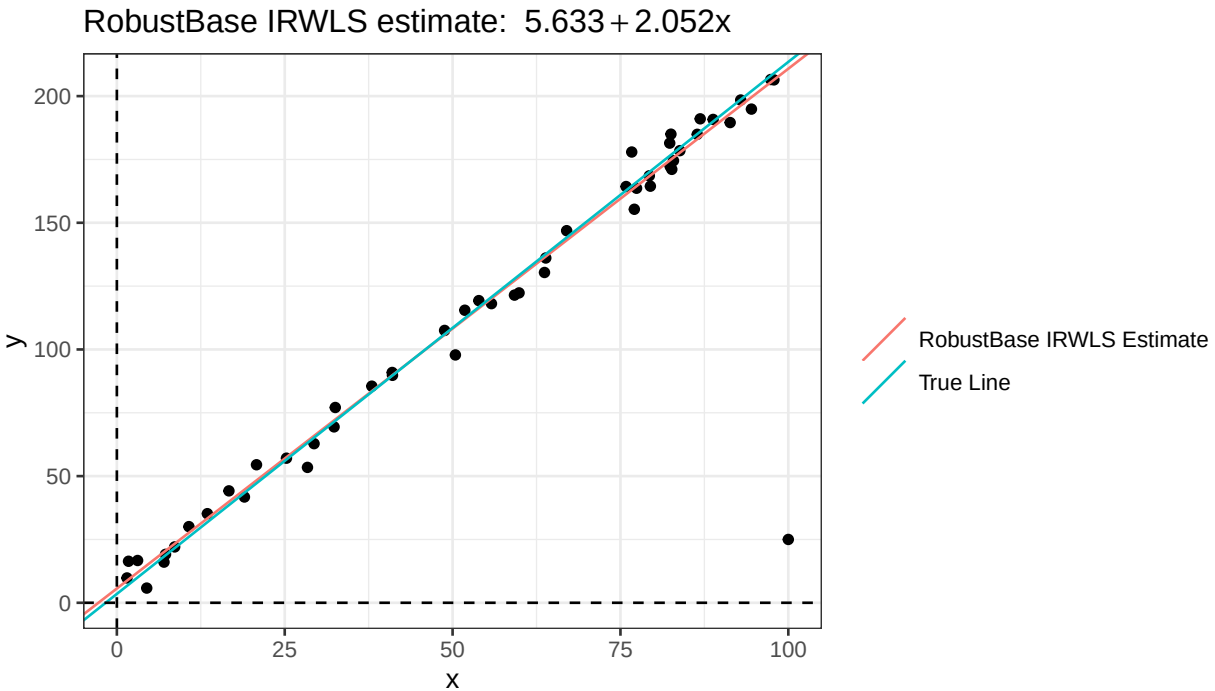


Figure 13: Scatterplot of the Data with the IRWLS Regression Line (robustbase package)

The β estimates are very close to the other weighted methods and fairly close to the true values, with the β_0 estimate of 5.633 (true value is 3.5) and β_1 estimate of 2.05 (true value is 2.1).

- (d) Fit a Quantile regression model based on x_2 and y_2 . Interpret the coefficients of the model; are they what you would expect?

```

quant_mod <- quantreg::rq(y ~ x, data = second_dat, tau = 0.5)
quant_sum <- data.frame(summary(quant_mod, se = "boot")$coefficients)

colnames(quant_sum) <- c("Estimate", "SE", "t", "p-value")
rownames(quant_sum) <- c("Intercept", "x")

quant_sum$'p-value' <- ifelse(quant_sum$'p-value' < 0.001, "<0.001", round(quant_sum$'p-value',3))

quant_sum <- xtable(quant_sum,
  label = "quant.model.sum"
, caption = "Quantile Model Summary (With Outlier)"
, digits = 3)

```

	Estimate	SE	t	p-value
Intercept	5.529	1.887	2.930	0.005
x	2.053	0.029	71.055	<0.001

Table 6: Quantile Model Summary (With Outlier)

```

ggplot(data = second_dat, aes(x = x, y = y)) +
  geom_point() +
  geom_abline(aes(intercept = quant_sum$Estimate[1],
                  slope = quant_sum$Estimate[2],
                  color = "Quantile Estimate")) +
  geom_abline(aes(intercept = 3.5,
                  slope = 2.1,
                  color = "True Line")) +
  theme_bw() +
  labs(color = "") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  ggtitle(bquote(
    "Quantile estimate: " ~
    .(round(quant_sum$Estimate[1], 3)) +
    .(round(quant_sum$Estimate[2], 3)) * x
  ))
))

```

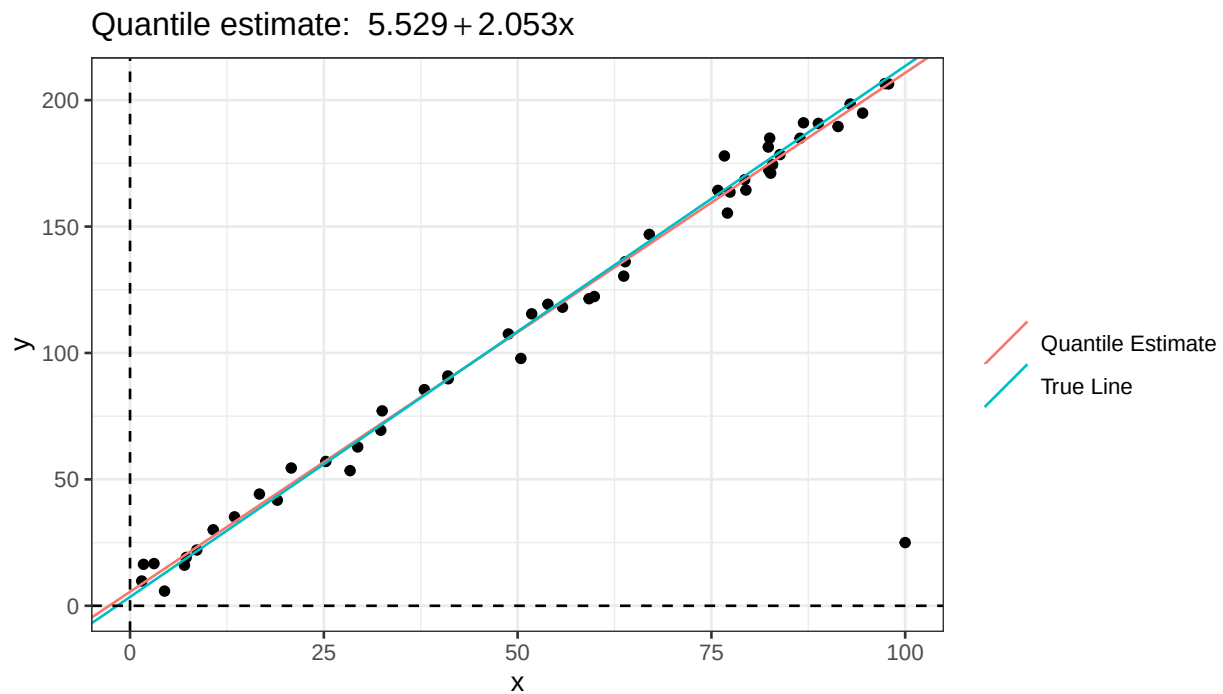


Figure 14: Scatterplot of the Data with the Quantile Regression Line

The estimates of the β 's for the quantile regression are very close to the true values: $\beta_0 = 5.529$ (true value is 3.5), $\beta_1 = 2.053$ (true value is 2.1). Visually, the regression lines are very similar as well.

4. **Optional, but recommended** Plot the results! Create a scatterplot of x_2 and y_2 . Superimpose the true regression line, the OLS estimate, the IRWLS estimates, and the quantile regression estimate. Interpret the results.


```

ggplot(data = second_dat, aes(x = x, y = y)) +
  geom_point() +
  geom_abline(aes(intercept = second_sum$Estimate[1],
    slope = second_sum$Estimate[2],
    color = "OLS")) +
  geom_abline(aes(intercept = IWH_sum$Estimate[1],
    slope = IWH_sum$Estimate[2],
    color = "Huber IRWLS")) +
  geom_abline(aes(intercept = IWB_sum$Estimate[1],
    slope = IWB_sum$Estimate[2],
    color = "Bisquare IRWLS")) +
  geom_abline(aes(intercept = rob_sum$Estimate[1],
    slope = rob_sum$Estimate[2],
    color = "RobustBase IRWLS")) +
  geom_abline(aes(intercept = quant_sum$Estimate[1],
    slope = quant_sum$Estimate[2],
    color = "Quantile")) +
  geom_abline(aes(intercept = 3.5,
    slope = 2.1,
    color = "True Line")) +

  theme_bw() +
  labs(color = "") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  ggtitle('Comparison of Model Regression Equations') +
  scale_color_manual(values = c("OLS" = "red",
    "Huber IRWLS" = "orange",
    "Bisquare IRWLS" = "yellow2",
    "RobustBase IRWLS" = "green3",
    "Quantile" = "blue",
    "True Line" = "purple3"),
    breaks = c("OLS",
    "Huber IRWLS",
    "Bisquare IRWLS",
    "RobustBase IRWLS",
    "Quantile",
    "True Line"))

```

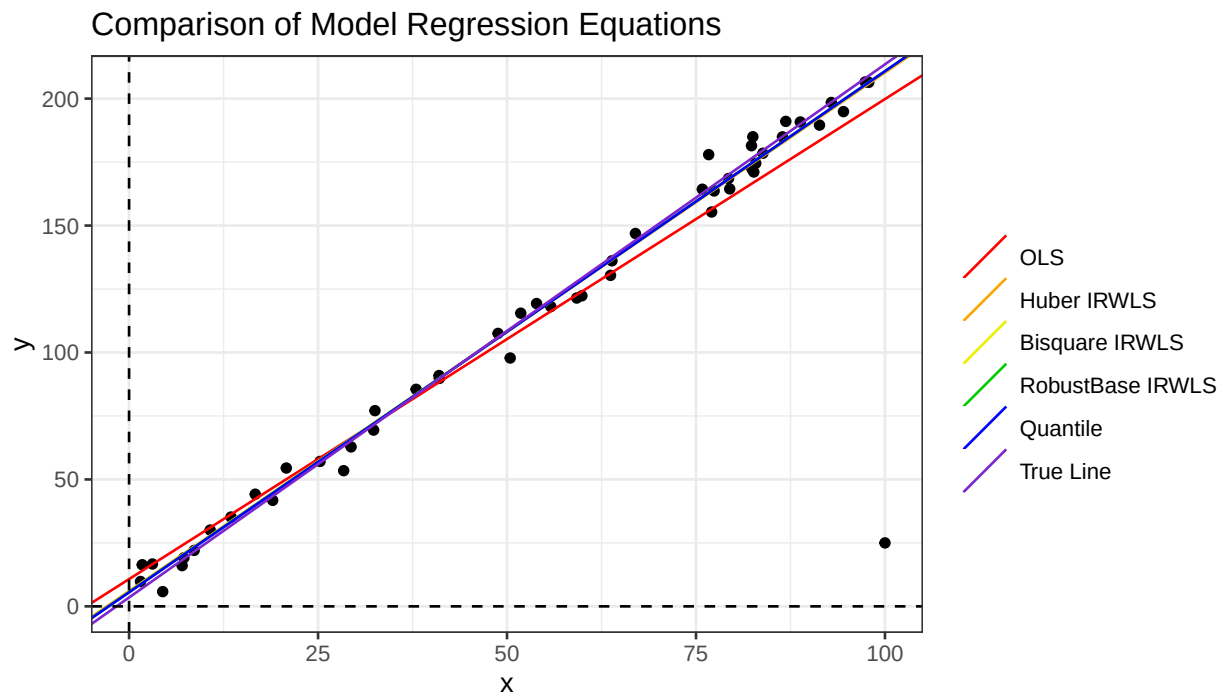


Figure 15: Scatterplot of the Data and Each Model

When comparing all of the models visually, the only one that stands out is the standard OLS model, with very little variation amongst the true line and the robust methods.

```

total_sum <- tibble(Method = c("OLS",
                              "Huber IRWLS",
                              "Bisquare IRWLS",
                              "RobustBase IRWLS",
                              "Quantile",
                              "True Values"),
  Intercept = c(second_sum$Estimate[1],
                IWH_sum$Estimate[1],
                IWB_sum$Estimate[1],
                rob_sum$Estimate[1],
                quant_sum$Estimate[1],
                3.5),
  Slope = c(second_sum$Estimate[2],
            IWH_sum$Estimate[2],
            IWB_sum$Estimate[2],
            rob_sum$Estimate[2],
            quant_sum$Estimate[2],
            2.1))

total_sum <- total_sum |>
  mutate("Intercept Error" = Intercept - total_sum$Intercept[6],
         "Slope Error" = Slope - total_sum$Slope[6])

total_sum <- total_sum[,c(1,2,4,3,5)]

total_sum <- xtable(total_sum,
  label = "total.model.sum",
  caption = "Comparison of Each Model",
  digits = 3)

```

Method	Intercept	Intercept Error	Slope	Slope Error
OLS	10.742	7.242	1.891	-0.209
Huber IRWLS	6.039	2.539	2.042	-0.058
Bisquare IRWLS	5.702	2.202	2.050	-0.050
RobustBase IRWLS	5.633	2.133	2.052	-0.048
Quantile	5.529	2.029	2.053	-0.047
True Values	3.500	0.000	2.100	0.000

Table 7: Comparison of Each Model

Likely coincidentally, each successive model we tested is closer to the true value than the last, with the biggest difference being after we moved from OLS to the robust methods.

5. **Optional, but recommended** Complete Questions 3-4, but use $n = 1000$. What changes? Why do you think that is?